

Table 4: Roles

Field Name	Data Type	Field Size	Description	Constraints
role_id	INT	4 bytes	Unique identifier for each role	PK, NOT NULL
role_name	VARCHAR	50 chars	Name of the role (e.g. Administrator, Department Head)	NOT NULL
position	VARCHAR	100 chars	Position title associated with the role	NULL

Manages user roles, user role, role id, access based upon role_id.

Table 5: Users

Field Name	Data Type	Field Size	Description	Constraints
user_id	INT	4 bytes	Unique identifier for each user	PK, NOT NULL
full_name	VARCHAR	150 chars	Full name of the user	NOT NULL
email	VARCHAR	150 chars	Login email address	UNIQUE, NOT NULL
password_hash	VARCHAR	255 chars	Encrypted password	NOT NULL
role_id	INT	4 bytes	Role assigned to the user	FK → Roles
department_id	INT	4 bytes	Department the user belongs to	FK → Departments
mobile_number	VARCHAR	20 chars	Mobile number for SMS notifications	NULL
is_active	BIT	1 bit	Account status: 1 = Active, 0 = Inactive	NOT NULL

User account details, login information and user department information for user authentication and access into the system

Table 6: Departments

Field	Data	Field	Descri	Const
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Name	Type	Size	ption	raints
department_id	INT	4 bytes	Unique identifier for each department	PK, NOT NULL
department_name	VARCHAR	100 chars	Full name of the department	NOT NULL

Manages store department details and also helps in document routing and users.

MODULE 2 — Document Management

Table 7: Document_Types

Field Name	Data Type	Field Size	Description	Constraints
type_id	INT	4 bytes	Unique identifier for document type	PK, NOT NULL
type_name	VARCHAR	100 chars	Name: Research Incentive, Travel Order, Leave Application, etc.	NOT NULL
arta_days	INT	4 bytes	Baseline processing days (3, 7, 20)	NOT NULL

Records the categories of the documents along with their associated workflows processes.

Table 8: Documents

Field Name	Data Type	Field Size	Description	Constraints
document_id	INT	4 bytes	Unique identifier	PK, NOT NULL
reference_number	VARCHAR	50 chars	System-generated Tracking ID	UNIQUE, NOT NULL
title	VARCHAR	300 chars	Document Subject / Title	NOT NULL
submitted_by	INT	4 bytes	User who register	FK (Users)

			ed the document	
departm ent_id	INT	4 bytes	Target routing departm ent	FK (Departm ents)
type_id	INT	4 bytes	Docum ent categor y classific ation	FK (Docume nt_Types)
attachm ent_pat h	VARC HAR	500 chars	Server path to the scanned file	NOT NULL
current_ step_in dex	INT	4 bytes	Current FSM stage (1 to 5)	NOT NULL
is_escal ated	BIT	1 bit	ARTA violatio n flag	NOT NULL, DEFAULT 0
date_fil ed	DATET IME	8 bytes	Timesta mp of registrat ion	NOT NULL

This table stores all the documents that are submitted and includes details about the documents such as document number, type of documents, which department, status of documents etc. This table acts as the primary table in the system.

MODULE 5 — Digital Signature + Audit Trail

Table 9: Digital_Signatures

Field Name	Data Type	Field Size	Description	Constraints
signature_id	INT	4 bytes	Unique identifier	PK, NOT NULL
document_id	INT	4 bytes	The document being signed	FK (Documents)
signed_by_user_id	INT	4 bytes	Official who provided signature	FK (Users)

			e	
signature_image	VARCHAR	500 chars	Base64/Path of visual signature	NOT NULL
signature_hash	VARCHAR	255 chars	SHA-256 integrity hash	NOT NULL
action_type	VARCHAR	50 chars	e.g., "Endorsed", "Approved"	NOT NULL
signed_at	DATETIME	8 bytes	Timestamp of signing event	NOT NULL

The Digital_Signatures table is where a cryptographic signature or attestation for every digital signature of a document is stored. It associates each signature file image with the SHA-256 hash that validates it.

Table 10: Audit_Trail

Field Name	Data Type	Field Size	Description	Constraints
audit_id	INT	4 bytes	Unique identifier	PK, NOT NULL
document_id	INT	4 bytes	The document being tracked	FK (Documents)
user_id	INT	4 bytes	User who performed the action	FK (Users)
action	VARCHAR	100 chars	Short action descriptor	NOT NULL
details	TEXT	Variable	Specific system logs	NOT NULL
timestamp	DATETIME	8 bytes	Exact time the action occurred	NOT NULL

			d	
Field Name	Data Type	Field Size	Description	Constraints

This Audit Trail is an immutable history book that records all actions that are taken on a document and who is performing that action. It is the record that serves as our back-stop and the record we use to audit, verify or settle a claim.

MODULE 6 — Monitoring Reports + Dashboard

Table 10: Report_Logs

Field Name	Data Type	Field Size	Description	Constraints
report_id	INT	4 bytes	Unique identifier for each report record	PK, NOT NULL
generated_by	INT	4 bytes	User who generated the report	FK → Users
report_type	VARCHAR	50 chars	Type: Document Summary, Department Performance, Workflow Report	NOT NULL
date_from	DATE	3 bytes	Start date of the report period	NOT NULL
date_to	DATE	3 bytes	End date of the report period	NOT NULL
generated_at	DATETIME	8 bytes	Date and time the report was generated	NOT NULL

ReportLogs The ReportLogs table keeps track of the reports being generated by authorised users including type and time period. It has a generated_by foreign key relating to the Users table. It allows reporting and auditing of generated reports.

MODULE 7 — Routing Slip

Table 11: Routing_Slips

Field Name	Data Type	Field Size	Description	Constraints
slip_id	INT	4 bytes	Unique identifier for each routing slip	PK, NOT NULL

document_id	INT	4 bytes	Document this routing slip is attached to	FK → Documents
tracking_number	VARCHAR	30 chars	System-generated tracking number assigned upon document receipt	UNIQUE, NOT NULL
received_by	INT	4 bytes	Receiving clerk who accepted and recorded the document	FK → Users
date_received	DATE TIME	8 bytes	Date and time the document was physically received	NOT NULL
sender_name	VARCHAR	150 chars	Name of the client or originating office that submitted the document	NOT NULL
action_instruction	TEXT	Variable	Instructions noted on the routing slip (e.g. For Signature, For Review)	NULL
target_department_id	INT	4 bytes	Department the document is being routed to	FK → Departments
noted_by	INT	4 bytes	Mayor or City Administrator who noted the routing slip	FK → Users
slip_status	VARCHAR	30 chars	Current status of the slip: Active,	NOT NULL

			Completed, Returned	
created_at	DATETIME	8 bytes	Date and time the routing slip was created in the system	NOT NULL

RoutingSlips The RoutingSlips table records the physical document circulation of documents that have arrived at the counter and records the name of the sender, the clerk who received the document, the procedure or instruction, the department which the document is targeted to. This table references the Documents table and Users and Departments tables in more than one place as a foreign key.

MODULE 8 — Smart Escalation (ARTA)

Table 12: ARTA_Escalations

Field Name	Data Type	Field Size	Description	Constraints
escalation_id	INT	4 bytes	Unique identifier for each escalation record	PK, NOT NULL
document_id	INT	4 bytes	Document that triggered the escalation	FK → Documents
Arta_threshold	INT	4 bytes	ARTA processing day limit breached: 3, 7, or 20 working days	NOT NULL
Days_elapsed	INT	4 bytes	Number of working days elapsed since document submission	NOT NULL
escalation_level	VARCHAR	20 chars	Severity level: Warning, Critical, Overdue	NOT NULL
Notified_user_id	INT	4 bytes	Oversight or compliance officer notified of the escalation	FK → Users
Notification_sent	BIT	1 bit	Whether the alert notification	NOT NULL

			n was sent: 1 = Sent, 0 = Pending	
resolved	BIT	1 bit	Whether the escalation has been resolved: 1 = Resolved, 0 = Unresolved	NOT NULL
escalated_at	DATETIME	8 bytes	Date and time the escalation was triggered	NOT NULL
resolved_at	DATETIME	8 bytes	Date and time the escalation was resolved	NULL